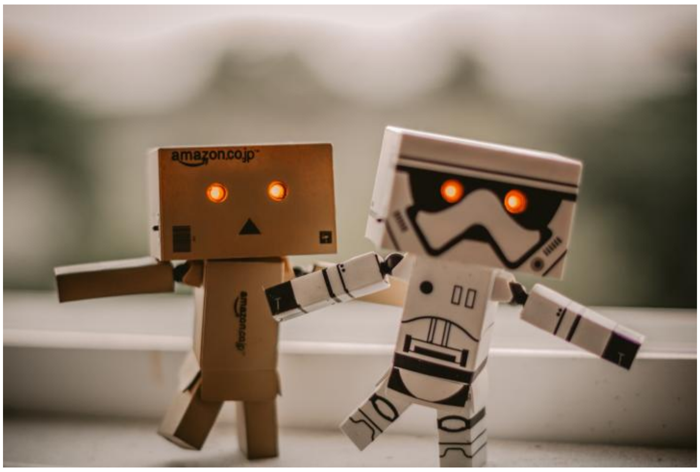

AI IN ROBOTICS

Coursework 1

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SUBMISSION DATE: 02 NOVEMBER 2025

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Introduction

The intelligent, ceiling-mounted office assistant Aerius autonomously identifies, organizes, and manages office supplies. Its name, inspired by 'air' and 'aerial' mobility, reflects its sleek design and ceiling-mounted operation. This report outlines Aerius's design, required skills, hardware and software architecture, dataset and implementation plan, safety measures, and budget considerations.

Mission and Tasks

Aerius maintains a clean, organized, and efficient workspace by locating, classifying, and storing items, while coordinating with other maintenance robots. It moves along a ceiling-mounted rail system, saving space and energy, and can be activated automatically or by command for tasks like inventory management, organization, and light cleaning. Its telescopic arm extends only as needed, using precise X, Y, Z positioning to handle objects safely.

Detailed Task List

1. Scanning and Mapping

- Uses LiDAR and cameras to scan the office environment.
- Creates and updates 3D maps of floors, desks, shelves, and drawers.
- Detects dirty or cluttered areas for cleaning coordination.

2. Item Detection

- Recognizes common office supplies via AI and computer vision.
- Differentiates between misplaced and categorized items.
- Detects dust or surface mess using floor sensors.

3. Priority Setting & Decision Making

- Evaluates workstation conditions to determine tasks (cleaning, sorting, tidying).
- Sets task priorities based on user preferences, item type, and urgency.
- Interfaces with office management system for planned maintenance.

4. Planning & Navigation

- Moves via ceiling-mounted rails with X, Y, Z axes for precise positioning.
- Stepper motors and sensors ensure safe, collision-free travel.
- Calculates shortest, safe routes and coordinates with cleaning robots.

5. Object Manipulation

- Equipped with multi-fingered, soft-grip end effector.
- Adjusts grip strength and angle according to object fragility and material.

6. Placement & Organization

- Automatically arranges desks, meeting tables, and common spaces.
- Sorts items into smart drawers or storage areas based on accessibility and usage.

7. Inventory & Monitoring

- Tracks identified objects in a digital catalogue.
- Assists in locating lost items accurately.

8. User Control & Management

- Responds to voice commands, gestures, or mobile app inputs.
- Provides feedback on task completion and inventory status.
- Supports custom scheduling for routine tasks.

9. Tidying & Cleaning

- Uses micro-vacuum or mini-cleaner to remove dust and debris.
- Floor sensors communicate with Aerius for cleaning schedules.

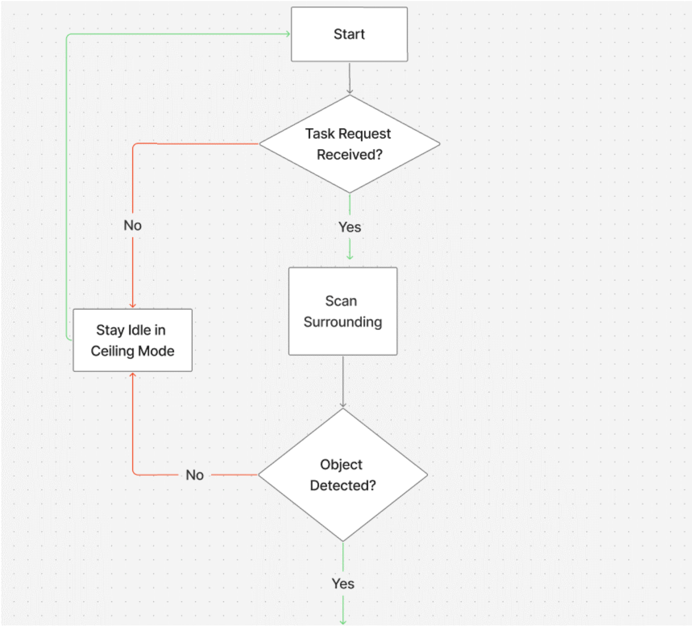
10. Docking & Self-Maintenance

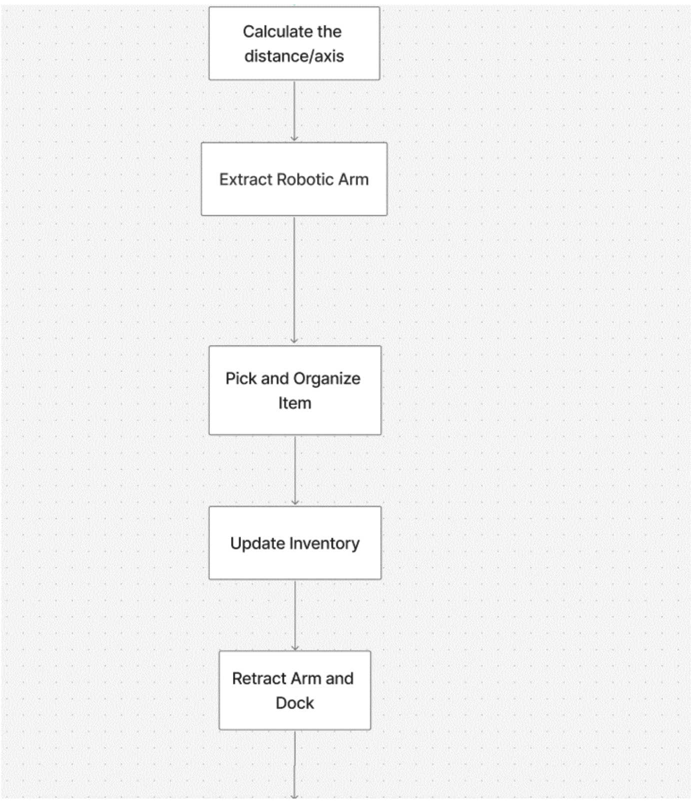
- Retracts to ceiling bay for wireless charging and self-cleaning.
- Runs diagnostics on sensors and grippers
- Updates firmware automatically via cloud.

11. Adaptive Learning

- Learns user behavior and workspace trends over time.
- Recognizes frequently used items to improve accessibility.

Below is a flowchart representation of the basic functions of Aerius:


Start
Task Request Received?
No
Yes
Scan Surrounding
Stay Idle in Ceiling Mode
Object Detected?
No
Yes


Calculate the distance/axis
Extract Robotic Arm
Pick and Organize Item
Update Inventory
Retract Arm and Dock

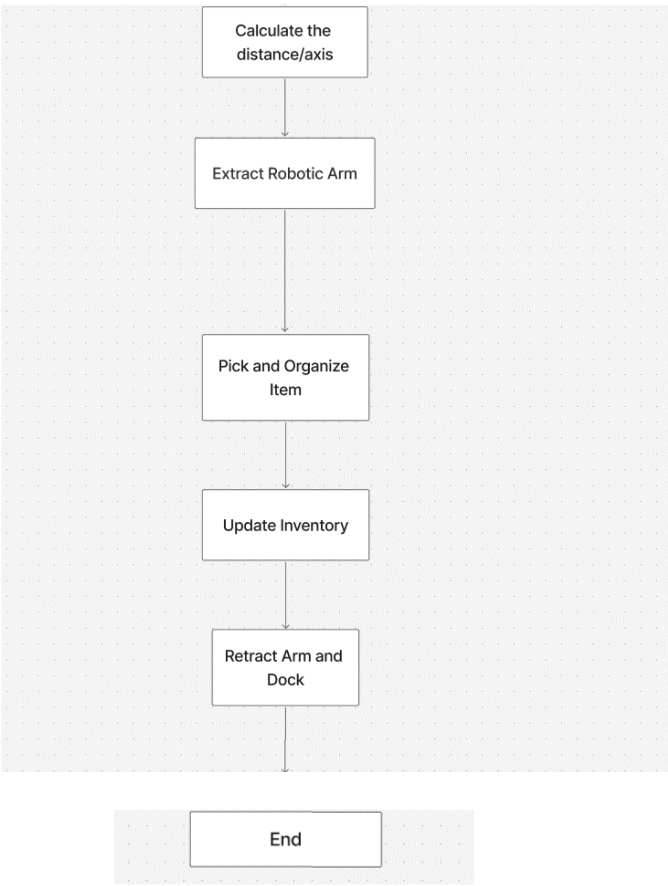


Figure 1 Fallback Decision flow

Required Skills

In the table below, Aerius's required skills to execute the tasks above, description of the input and output criteria, along with the success criteria are explained.

Skills and Description	Interface	Input Criteria	Output Criteria	Success Criteria
Perception and Object Recognition: Detects office supplies, obstacles, clutter, and areas needing cleaning.	RGB + Depth Cameras: Capture images and depth data simultaneously to recognize items, shapes, and clutter. LiDAR Sensors: Generate 3D spatial maps and calculate distances to obstacles; 3D point cloud processing helps accurately arrange objects. Ultrasonic/Proximity Sensors: Detect nearby obstacles, humans, or furniture edges; CNNs classify objects (e.g., pen vs. ruler). Multi-Sensor Fusion: Integrates data from multiple sensors to identify, locate, and track targets. YOLOv8: algorithm providing bounding boxes and supporting segmentation and pose estimation. (Wang. et al, 2023)	Sound-wave reflections from proximity sensors for short-range obstacle detection and safe navigation. Multi-sensor fusion combining data to improve robustness against noise and errors. RGB images for YOLOv8 to detect and localize objects in real-time via bounding boxes. Motion detection and lighting condition data. Pre-trained neural network weights for item recognition.	Type, position, size, and orientation of recognized items. Identification of untidy, messy, or unclean areas. Obstacle maps for navigation and task planning. Alerts for abnormalities or unknown objects. Coordinates of obstacles and humans for safe path planning.	The robot is placed in a simulated office environment with various objects and clutter. It must correctly identify item type, size, and orientation with ≥90% accuracy, detect messy or unclean areas with ≥85% accuracy, generate obstacle maps with ≥95% completeness, issue alerts for unknown objects, and provide obstacle and human coordinates with ≤10 cm positional error for safe navigation.

Figure 2 Required Skills (1)

Skills and Description	Interface	Input Criteria	Output Criteria	Success Criteria
Localization and Mapping: Keeps spatial awareness to navigate the arm module safely.	Ceiling Rail System: Encoders and servo motors on X, Y, Z axes allow movement along rails. SLAM enables creation and updating of a dynamic 3D map. RGB + Depth Cameras: Capture color and depth images for 3D environmental mapping. Path Planning (SI-ACO) (Zhang et al, 2023): Self-Improved Ant Colony algorithm fuses camera and LiDAR data to compute shortest paths to targets. Motion Estimation: IMU and encoder data processed via Kalman Filtering for accurate position tracking. Coordinate Transformation: Rail movements converted into 3D workspace coordinates for precise object interaction. ROS Communication: Navigation, control, and	LiDAR & Depth Sensors: Scan the environment to generate spatial maps. Floor Sensors: Provide motion, cleanliness, and occupancy information. Object Recognition Module: Updates item positions in real time. Task Requests: Commands such as 'clean desk' or 'organize desk.' IMU Data: Acceleration and angular velocity for motion estimation. Motion Correction: LiDAR data corrects predicted motion by comparing measured distances with previously mapped features, reducing drift. Path Planning: Combines sensor data to compute the shortest and safest route to targets.	Real-time representation of floors, walkways, shelves, and desks. Tracks manipulator location relative to obstacles and objects. Alerts for blocked paths or layout changes. Communicates with cleaning robots to prevent task overlap. Displays messages such as 'Path complete' or 'Obstacle detected' on the dashboard.	The system operates in a mapped workspace with dynamic elements. It must generate a real-time layout of floors and furniture with ≥95% accuracy, track the manipulator's position relative to obstacles within ≤5 cm error, issue blockage or layout change alerts, coordinate tasks with cleaning robots with zero overlap, and display clear status messages like 'Path complete' or 'Obstacle detected' on the dashboard in real time.

Figure 3 Required Skills (2)

	RGB + Depth Cameras: cleaning modules communicate through the Robot Operating System. Sensor Fusion: Multi-sensor module integrates IMU, camera, LiDAR, and encoder inputs to stabilize pose estimation. Loop Closure: Revisiting mapped areas triggers loop closure detection, correcting accumulated localization errors and aligning new and existing map segments.			

Figure 4 Required Skills (3)

Skills and Description	Interface	Input Criteria	Output Criteria	Success Criteria
Organizing and scheduling: decides on the best order for tasks and carries out planned actions.	Inventory Database API: Provides object location, type, and usage frequency; supports A* path planning for optimal routes. Task Scheduler / Planner Interface: Retrieves user-defined schedules and priorities for task execution. User App / Voice Control API: Accepts commands and schedules from employees. Robot Operating System (ROS) / Control Software: Generates optimized work plans by integrating perception and localization data. PID Motor Control: Ensures smooth acceleration, deceleration, and precise movement along all axes.	Information on object locations, usage frequency, and task priorities. Cleaning or tidying triggers from floor corner/edge sensors. Commands, schedules, and preferences from employees. Real-time state of cleaning robots and ceiling arm.	Real-time 3D office map including floors, walkways, shelves, and desks. Location of the arm and other robots relative to obstacles and objects. Identified items with type, position, size, orientation, and confidence scores. Marked untidy, messy, or unclean areas. Navigation and path-planning updates, including dynamic warnings for blocked routes or layout changes. Task status and notifications: e.g., cleaning triggers, task assignments, planned sequences, and conflict resolution between robots.	The system generates a real-time 3D office map showing floors, shelves, and walkways with ≥95% mapping accuracy. It tracks the arm and robots' positions within ≤5 cm deviation from true coordinates. Detected items must include type, position, size, and orientation with ≥90% confidence. Untidy or dirty areas are correctly flagged in ≥90% of trials. Path-planning updates must respond to blocked routes, and the system should display accurate task statuses, cleaning triggers, and conflict resolution messages on the dashboard in real time.

Figure 5 Required Skills (4)

Skills and Description	Interface	Input Criteria	Output Criteria	Success Criteria
Arm control and manipulation: Regulates a telescopic arm to pick and position objects safely.	Telescopic Robotic Arm: Extends and retracts to pick or place objects. Uses Inverse Kinematics to calculate joint angles for precise 3D positioning. Multi-Fingered End Effector with Force Sensors: Adapts grip based on object fragility and weight. Force Control Algorithm dynamically adjusts grip strength. Arm Control Software / Motion Planning API: Plans collision-free trajectories. Collision Detection & Recovery stops motion if unexpected resistance or obstruction occurs. Visual Servoing: Eye-to-hand cameras capture images for servoing and fiducial recognition. AprilTag is used to detect object bounding boxes. (Li et al, 2024). Proximity & Edge Sensors: Ensures arm avoids staff or nearby objects. Activates	Location, weight, size, and type of the object. Limits set by the localization and workspace module. Real-time detection of obstacles and human presence.	Commands for grip adjustment and arm trajectory execution. Placement of objects in correct workspace or storage locations. Retraction of the arm upon task completion. Status feedback indicating success or failure of actions.	The arm executes grip and trajectory commands accurately, positioning objects within ≤1 cm of the target location. Objects are placed in correct storage areas or workspaces ≥95% of the time. The arm retracts smoothly after completing tasks, and status feedback is correctly displayed or signaled in 100% of test cases.

Figure 6 Required Skills (5)

	dynamic avoidance when obstacles or people are detected. Dynamic Obstacle Avoidance (DWA): Recalculates motion paths in real-time to avoid moving obstacles.			

Figure 7 Required Skills (6)

Skills and Description	Interface	Input Criteria	Output Criteria	Success Criteria
Human Robot Interaction and Voice Command: Processes employee requests via speech, gestures, or apps. Schedules and executes tasks accordingly.	Microphone Array / Noise Filtering: Captures and filters user voice commands. Gesture Detection Camera / Sensors: Recognizes basic human gestures using pose estimation frameworks (e.g., MediaPipe, OpenPose). Mobile App / Web Interface: Allows remote task scheduling and command input; queued commands are prioritized by the Task Scheduler Algorithm. NLP / Speech Recognition: Converts spoken instructions into structured tasks. Intent Classification Models (e.g., BERT): Analyze user context and requests for accurate task interpretation.	Voice commands captured via the microphone array. Human gestures detected using motion sensors and cameras. Task plans, user preferences, and scheduling from the mobile/web app. User identification and contextual information for personalized interaction. Feedback from perception and localization modules (e.g., obstacles, task status).	Confirmation of task completion or schedule updates. Suggestions for reorganization, cleaning, or inventory adjustments. Alerts if a task cannot be completed or if conflicts arise. Visual or auditory feedback confirming command reception and execution. Automated execution of repetitive tasks via the scheduling system.	The system accurately executes voice, gesture, or app commands with at least 98% success, provides visual or auditory confirmation of receipt and completion, issues alerts for uncompleted tasks or conflicts, and automates repetitive tasks according to schedule ≥95% of the time.

Figure 8 Required Skills (7)

Skills and Description	Interface	Input Criteria	Output Criteria	Success Criteria
Inventory and Cleaning Management: Tracks items, updates inventory, and schedules cleaning using floor corner sensors.	Inventory Database API: Updates object status and locations in real time via cloud-synced logs. Floor Corner & Edge Sensors: Detect dirt, debris, and congested zones; Edge Detection Filter identifies particles near corners. Dirt Prediction Model: ML-based model forecasts high-use areas for prioritized cleaning. User App / Notifications API: Alerts users about untidy zones or missing items; Change Detection Algorithm compares live camera feed with inventory map. IoT Communication: MQTT protocol enables coordination with dashboard and other systems. RFID Tracking: Continuous monitoring of tagged office assets.	Edge & Floor Corner Sensors: Detect lost items, dirt, or debris. Recognizes objects, providing location, size, and type. Tracks current status of stored and frequently used items. Confirms completed cleaning tasks and identifies areas needing attention.	Real-time updates of item locations, newly added items, or missing objects in the database. Commands sent to floor-cleaning robots or signals for human assistance. Alerts employees about missing, misplaced, or unclean items.	The system updates the inventory in real time with at least 98% accuracy, sends appropriate commands to cleaning robots or humans, notifies employees about missing or misplaced items, achieving ≥95% successful execution.

Figure 9 Required Skills (8)

Hardware Components

Sensors

1. RGB Camera

Selected: Intel RealSense D415 - It combines RGB and depth sensing, offers an 87° field of view, a depth ranges up to 6 m, USB connectivity, and native ROS2 & OpenCV support, making it suitable for precise measurements in various lighting conditions (Intel, 2025).

Alternative: LUCID Helios2 Wide - Provides a wider field of view (108° × 78°) and depth sensing up to 8.3 m, suitable for ceiling-mounted applications, but lacks an integrated RGB module. (LUCID, 2025).

2. Depth Camera / RGB-D

Selected: OAK-D (Luxonis) - Compact and ideal for robotic integration, it combines stereo depth with onboard AI inference, providing good depth accuracy from 70 cm to 12 m with less than 2% error below 4 m, while reducing processing load (Luxonis, 2025).

Alternative: Intel RealSense D435 - Suitable for low-light conditions and budget-friendly, with a strong external compute requirement, but has a narrower field of view (65° × 40° in HD mode) (Intel, 2025).

3. 2D LiDAR

Selected: Slamtec RPLIDAR S2 - Offers a 30 m measuring range, 32 k measurements/sec sampling, 0.1125° angular resolution, and is dust- and waterproof, making it highly accurate and durable (SLAMTEC, 2025).

Alternative: RPLIDAR A3 - Suitable for indoor mapping and dynamic obstacle detection, with precise and fast scanning at a more budget-friendly price (SLAMTEC, 2025).

4. IMU (Inertial Measurement Unit)

Selected: Bosch BNO055 - Integrates onboard sensor fusion for precise orientation and motion tracking, reducing software complexity (Bosch Sensortec, 2025).

Alternative: MPU-9250 - Offers a 9-axis sensor at lower cost, but requires additional software to perform sensor fusion (InvenSense, 2025).

5. AprilTags / Fiducial Markers

Selected: AprilTag v2 - Provides highly accurate localization and mapping for robotics with minimal computational load (Luxonis, 2025).

Alternative: ArUco Markers - Commonly used in computer vision and easily integrated via OpenCV, but slightly less precise (Luxonis, 2025).

6. Safety Proximity Sensors

Selected: HC-SR04 Ultrasonic Sensor - Cost-effective and reliable for short-range distance measurements, suitable for preventing collisions in close-quarters tasks (SparkFun, 2023).

Alternative: VL53L1X Time-of-Flight (ToF) Sensor - Offers faster sampling and higher accuracy, enhancing obstacle detection in dynamic environments (Microelectronics STM, 2022).

Manipulations

1. **Telescopic Robotic Arm**

Selected: UFactory xArm 6 DoF - Provides precise 6-DOF movement for confined spaces, integrates easily with ROS, and has sufficient payload capacity with high-performance brushless servo motors (UFactory, 2025).

Alternative: Dobot Magician - A more affordable 4-DOF robotic arm suitable for educational purposes, but limited in precision and workspace flexibility compared to the xArm (Dobot, 2025).

Note: To cover a larger working range, we can add the Electric Linear Actuator.

2. **Arm Linear Actuator**

Selected: Electric Linear Actuator - Allows precise and controlled extension/retraction of the telescopic arm when integrated with ROS-based motion planning (Firgelli Automations, 2023).

Alternative: Pneumatic Cylinder - Offers faster movement and higher load capacity but requires compressors and more maintenance (SMC Corporation, 2023).

3. **Soft-Grip End Effector**

Selected: Robotiq 2F-85 Adaptive Gripper - Versatile for handling various objects with shape adaptation without causing damage (Robotiq, 2025).

Alternative: OnRobot RG2 - Compact design suitable for lightweight objects with adequate payload capacity (OnRobot, 2025).

4. **Ceiling Rail Movement Actuators**

Selected: Stepper motor linear gantry system - Provides precise and repeatable positioning along ceiling rails, ideal for accurate ceiling-mounted motion (Oriental Motor, 2025).

Alternative: Servo-driven linear actuators - Offer high torque and speed but require more complex control systems and are costlier (SMC, 2025).

5. **Rail System Controller / Motion Controller**

Selected: Stepper motor driver stack - Trinamic TMC-series drivers on a dedicated controller provide silent, smooth, and precise ceiling gantry motion with microstepping and stall-detection, balancing performance and cost (Trinamic Motion Control, 2023).

Alternative: Servo motor drive - Delta/Kollmorgen closed-loop drives deliver higher speed and torque for heavier payloads but require more complex tuning and safety measures (Siemens, 2023).

6. **Ceiling Mount Frame / Docking Bay**

Selected: Custom aluminum frame (CNC-milled) - provides a robust, vibration-resistant structure for secure ceiling-mounted operation (Bosch Rexroth, 2023).

Alternative: Modular 3D-printed mount (PLA/ABS) - allows rapid prototyping and cost-effective customization but may lack the rigidity of metal frames (Ultimaker, 2023).

Communication

1. **Microphone Array**

Selected: ReSpeaker 6-Mic Array - provides reliable voice command detection with beamforming and noise suppression for accurate recognition (ReSpeaker, 2023).

Alternative: Logitech Mic Array - more affordable and sensitive but offers lower directional accuracy (Logitech, 2023).

2. **Visual Feedback Display (Dock / Robot)**

Selected: 2.4" color TFT display + RGB LED ring - displays text, icons, and progress bars with immediate, clear status feedback when combined with an LED ring (Adafruit, 2023).

Alternative: Single-color LED matrix - more affordable and visible, but less flexible for conveying complex status indicators (Waveshare, 2023).

Cleaning

1. **Dust Filter / HEPA Module (for micro-vacuum)**

Selected: Replaceable HEPA13 cartridge + washable prefilter - ensures indoor air quality by efficiently capturing fine particles and allowing easy replacement (Honeywell, 2023).

Alternative: Electrostatic precipitator + washable filter - can reduce airflow resistance and capture particles but requires more maintenance and is sensitive to moisture (Dyson, 2023).

2. **Sensor Cleaning Mechanism (Air-jet / Wiper)**

Selected: Small solenoid-controlled air-jet + silicone wiper blade - provides an affordable and efficient way to remove loose dust from cameras and lenses in the field (DustTech, 2023).

Alternative: Ultrasonic vibration dust removal / electrostatic coating - reduces mechanical wear but is more costly and complex to implement (Nano-Clean, 2023).

Safety

1. **Emergency Stop Switch / Safety Cutoff**

Selected: Schneider Electric XB4 Series - industrial-grade emergency stop providing rapid power isolation and reliable operator safety (Schneider Electric, 2023).

Alternative: Omron A165E Series - more compact and suitable for ceiling installations, but may respond slower in some cases (Omron, 2023).

2. **Safety Bump / Proximity Sensors (Arm & Rail)**

Selected: Multi-zone time-of-flight (ToF) proximity sensors + capacitive touch strips - provide fast, adjustable, non-contact detection and immediate stopping on contact for enhanced human safety (STMicroelectronics, 2023; Microchip, 2023).

Alternative: Mechanical bump sensors and simple IR proximity - cost-effective and easy to implement but less reliable at close range and sensitive to ambient light (SparkFun, 2023; Sharp, 2023).

Software Architecture

Aerious software follows a modular ROS 2 architecture, where sensor nodes supply raw data to perception and localization modules. These modules process the environment and provide actionable information to the planning and control nodes. Consider the diagram below showing the dataflow in Aerious' system.

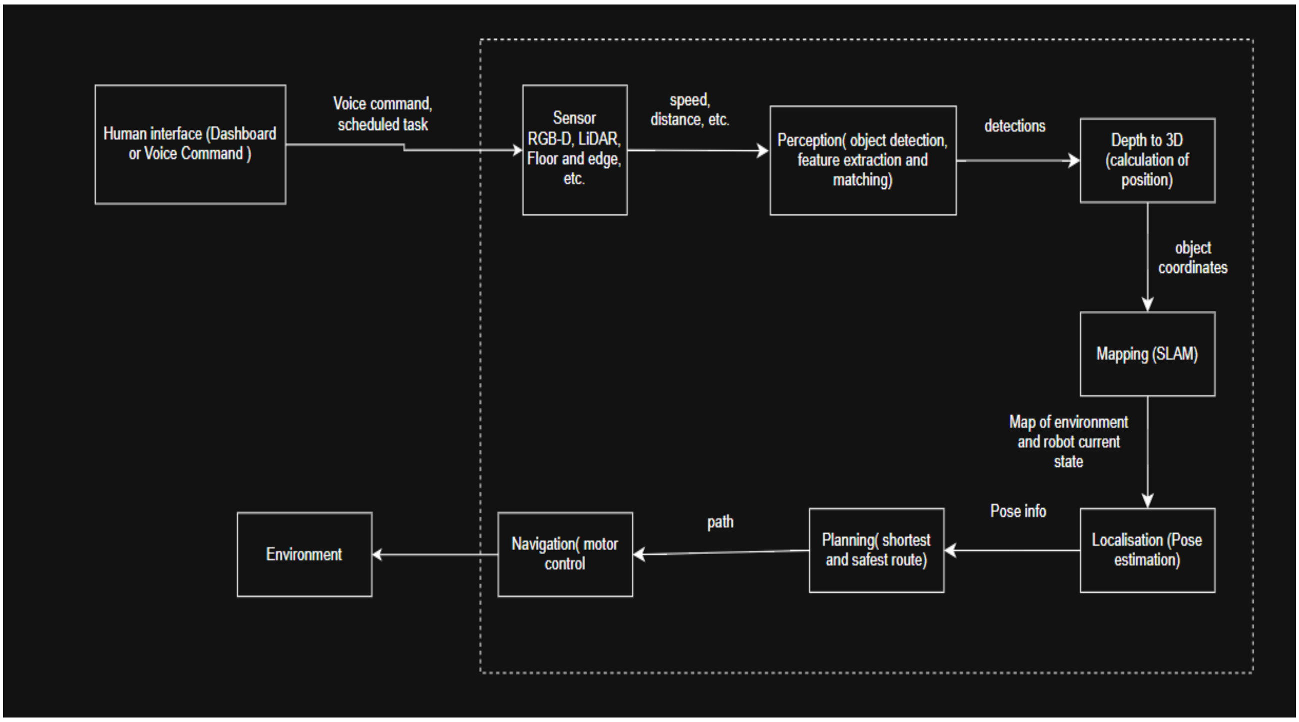


Figure 10- Module communication flowchart

ROS TOPICS AND FRAMES

Module	Example ROS 2 Topic	Data Type/ Message	Frame
Sensors	/rgb_image_topic /imu_data, /lidar, /floor, /proximity, /fused_map	/sensor_msgs/image, sensor_msg/laser, sensor_msg/imu	Camera_link, Base_link
Perception	/detected_objects	Vision_msg/detection3darr	Camer_link
Mapping	/map	nav_msg/occupancyGrid	Map
Localization	/odom	nav_msg/odometry	Odom
Planning	/path	nav_msgs/path	Map
Navigation	/cmd_vel	geometry_msg/grasp	Base_link

Figure 11 ROS Topics and Frames

The ROS2 dataflow diagram shows the interaction between the robot's perception, planning, and control modules. Sensor nodes publish data to topics, which the Sensor Fusion Node combines. The fused data is processed by the YOLOv8 detection node, then sent to the SI-ACO path planner to generate an optimized route. Motion Control converts this path into velocity commands, while safety nodes monitor obstacles and trigger emergency stops when necessary.

Consider this diagram below.

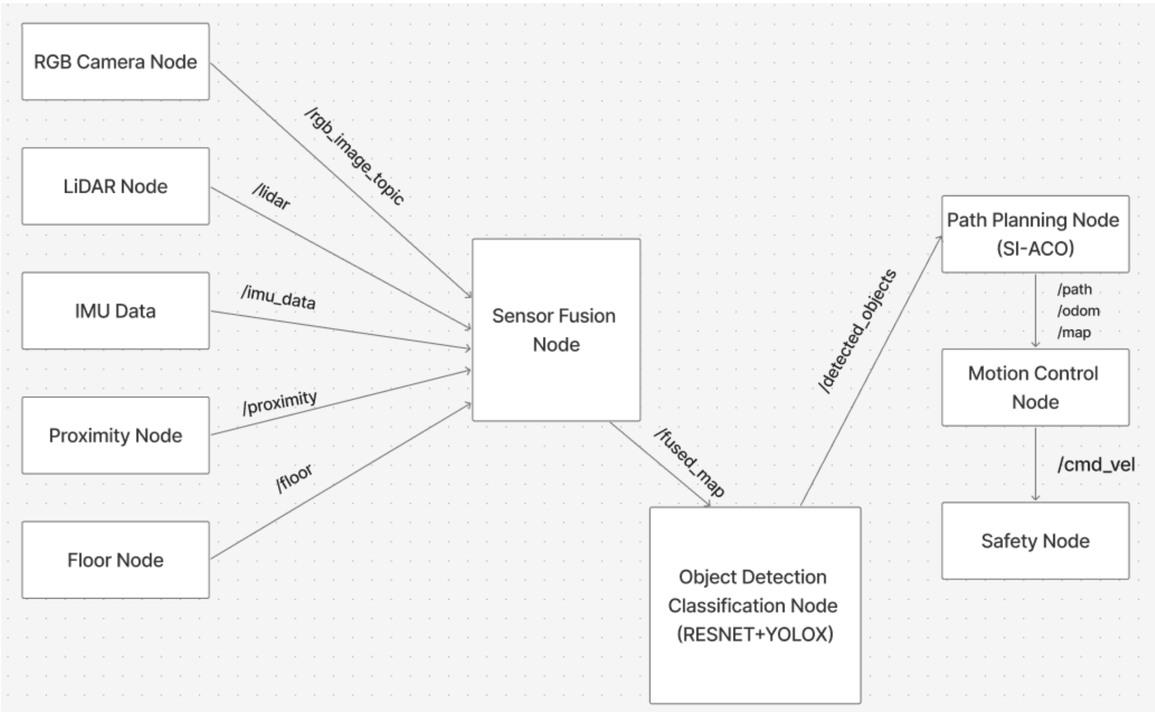


Figure 12- Robot Interaction

Dataset and Model Plan

The image classification model was created to recognize a total of 10 classes of office items: notebook, backpack, pen, file cabinet, mug, trashcan, laptop, chair, table and mouse.

An online Office-Home dataset (Venkateswara et al., 2017) of 62 office item classes was obtained for training and validating the model. The dataset was organized in 4 major folders named Clipart, Art, Product and Real World and each folder contained subfolders containing specific office items. Folders that contained office items not related to the chosen classes were removed. The dataset was organized using a python script to merge the same office items across the 4 major folders into one distinct folder. The names of the folders containing a particular office item served as a class label for that item and each class contains around 300 images. Some unclear and cropped images were removed and replaced by other photos online found online. An additional dataset was obtained from another online site (Images.cv, 2025) to test the model. These datasets were chosen because they had images pertaining to the chosen classes, they had a variety of images and a satisfactory amount of images needed for transfer learning.

Transfer learning was carried out using ResNet34, a 34-layer convolutional neural network selected for its ability to capture complex patterns and outperform shallower models like ResNet18 in accuracy (Subaar et al., 2024). Additionally, ResNet34 achieves high classification accuracy across various domains (Subaar et al., 2024) when leveraging pretrained ImageNet weights. It requires a relatively small dataset, which suited our scenario with roughly 3,000 training images, and it’s less prone to overfitting than ResNet50. Unlike ResNet50, ResNet34 demands less computational power, making it compatible with the limited resources of our devices. Overall, ResNet34 was an ideal choice for developing an office classification model, as it offers a good balance between accuracy and efficiency in resource-limited settings.

To customize ResNeT34 to the office item domain, several parameters were altered. The optimization function was changed from Adaptive Moment Estimation (Adam) to Stochastic Gradient Descent (SGD) because SGD performs better than Adam in terms of better generalization on image classification tasks (Gupta et al., 2021). During training, different learning rates were used when training the classifier head and when fine-tuning the entire model. The changes in learning rates were necessary because the final classification layer is randomly initialized each run, affecting the model's sensitivity to the learning rate.

The loss function that was used was cross-entropy loss because it is suitable for multi-class classification. It guides models to output probabilities close to true class labels by minimizing errors between predicted and actual outcomes. Cross-entropy loss was therefore, suitable in the office item classification scenario.

To find the appropriate learning rates, a learning rate finder was used to plot loss against learning rate. The value chosen for the head of the model was chosen around the "valley" before the loss increases sharply.

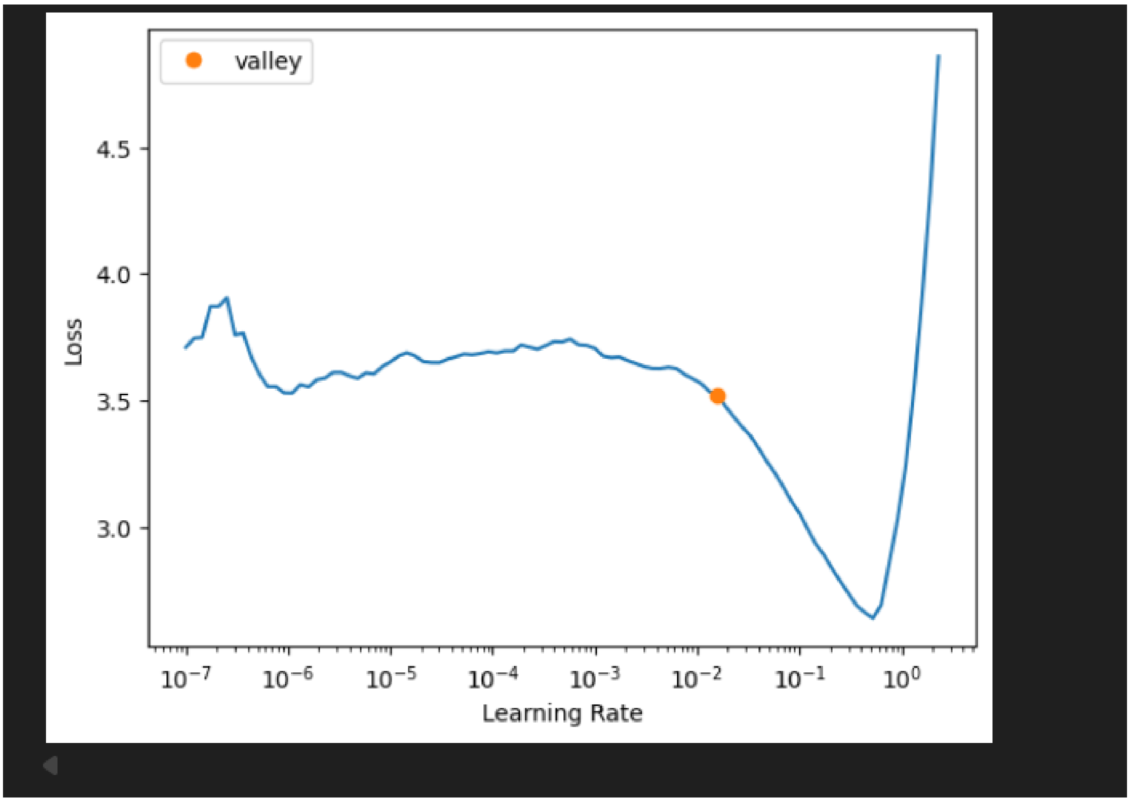


Figure 13: Learning rate graph

Based on figure 4, the learning rate of 1e-2 was used for training the head because the valley mark is around 1e-2.

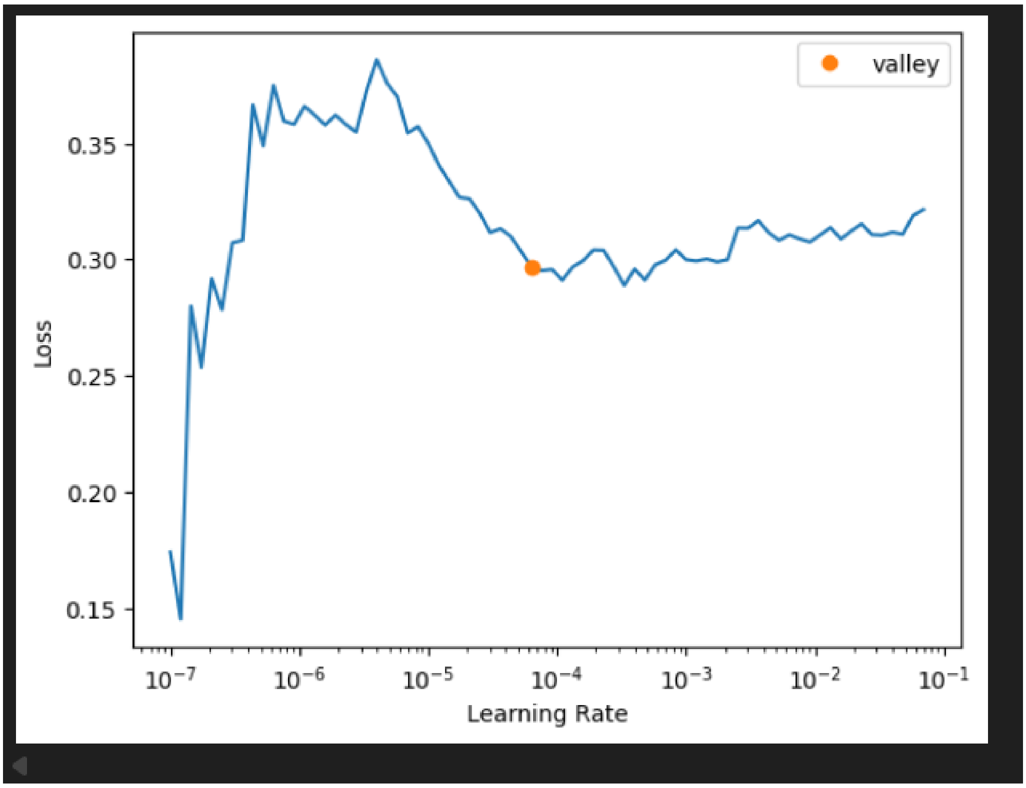


Figure 14: Second learning rate graph

In Figure 5, the loss is steady from the learning rate of 1e-5 to 1e-4, where the valley mark is. This range, 1e-5 to 1e-4, was used because that is the region where the model learns efficiently as the loss decreases gradually. Therefore, this range was chosen for fine-tuning the entire model to ensure that the pretrained weights were not disrupted.

For performance evaluation, accuracy, recall, precision, F1 score, macroF1 and confusion matrix calculated.

Error Analysis

This section of the report analyses the training results and performance of the model on a test dataset and discusses the error.

Recall was used to measure the ability of the model to correctly identify items of a given class and to ensure that the model does not miss or overlook some office items.

Precision was used to evaluate the model on how reliable it is in predicting a certain class and not confusing it for another class.

F1 Score was also used to balance precision and recall preventing bias and ensure fair evaluation.

A confusion matrix was also used to visually show classes the model correctly and incorrectly categorized to help identify where misclassification occurred.

Finally, accuracy was also used to show how well the model can correctly categorize an office item.

epoch	train_loss	valid_loss	accuracy	time
0	2.477132	0.678465	0.822414	06:06
1	1.092263	0.218193	0.937931	07:05
2	0.560991	0.162109	0.948276	06:25
3	0.394233	0.148353	0.956897	06:54
4	0.334851	0.140473	0.962069	06:37

Figure 15: Head training

Figure 6 shows the results of the head training. There were no signs of overfitting during the head training phase. Overfitting occurs when the training loss decreases but the validation loss increases, meaning it performs well on the training data but poorly on the unseen validation data it decreased.

In this case, both the training loss and validation loss kept on decreasing with each epoch from 0.678 on epoch 0 to 0.140 on epoch 4 with an accuracy of 96.2%, this shows that the model is generalising effectively and learning features of the distinct classes.

epoch	train_loss	valid_loss	accuracy	time
0	0.319290	0.139288	0.958621	06:26
1	0.289641	0.137094	0.962069	29:18
2	0.300452	0.143470	0.955172	06:15
3	0.321374	0.134729	0.958621	06:09
4	0.328110	0.137803	0.963793	06:20
5	0.319247	0.144360	0.953448	08:56

Figure 16 Entire Modelling training

Figure 7 shows results of the entire model training. There were signs of minor overfitting starting after epoch 1 with the validation loss increasing from 0.137 to 0.143 in epoch 2, but after the increase the model reaches its lowest validation loss of 0.134 in epoch 3 which showed better generalisation at that point. At epoch 5 is where the validation loss is at its highest among the epochs at 0.144, it is at this point where there is a clear sign of overfitting.

Detailed Classification Report:

	precision	recall	f1-score	support
Backpack	0.91	1.00	0.95	40
Chair	1.00	1.00	1.00	44
File_Cabinet	0.82	1.00	0.90	36
Laptop	1.00	1.00	1.00	47
Mouse	1.00	0.98	0.99	48
Mug	0.98	1.00	0.99	40
Notebook	1.00	0.97	0.99	36
Pen	1.00	1.00	1.00	51
Table	1.00	0.87	0.93	38
Trash_Can	1.00	0.84	0.91	37
accuracy			0.97	417
macro avg	0.97	0.97	0.97	417
weighted avg	0.97	0.97	0.97	417

Figure 17 Classification Report

Figure 8 shows a classification report of the model. F1 score, macro-F1, accuracy, recall, precision were used to analyze the model. The range of precision across all classes is from 0.82 being file cabinet to 1 which include. File cabinet having a precision of 0.82 can be due to the similarities in features like the drawers it has with an office table/desk. For classes with a precision of 1 like chair, pen and table means that it is very unlikely for the model to confuse the classes for another class whereas for a class like file cabinet, the model can misclassify file cabinet around 20% of the time. The range of recall across classes is from 0.84 to 1, with 0.84 being trash can. This means around 16% of the time the model could not correctly identify a trash can, this could have been due to the different varieties, shape or even features of a trash can. The f1 score ranges from 0.91 to 1, with 0.91 being trash can. The range indicates that there are low false positives and low false negatives meaning there is a high probability of high correct predictions and rarely misses an instance of a class, therefore there is a balance between precision and recall. The macro F1 score is 0.97, which indicates that the model is good at identifying a particular class and rarely mistakes a class for another class.

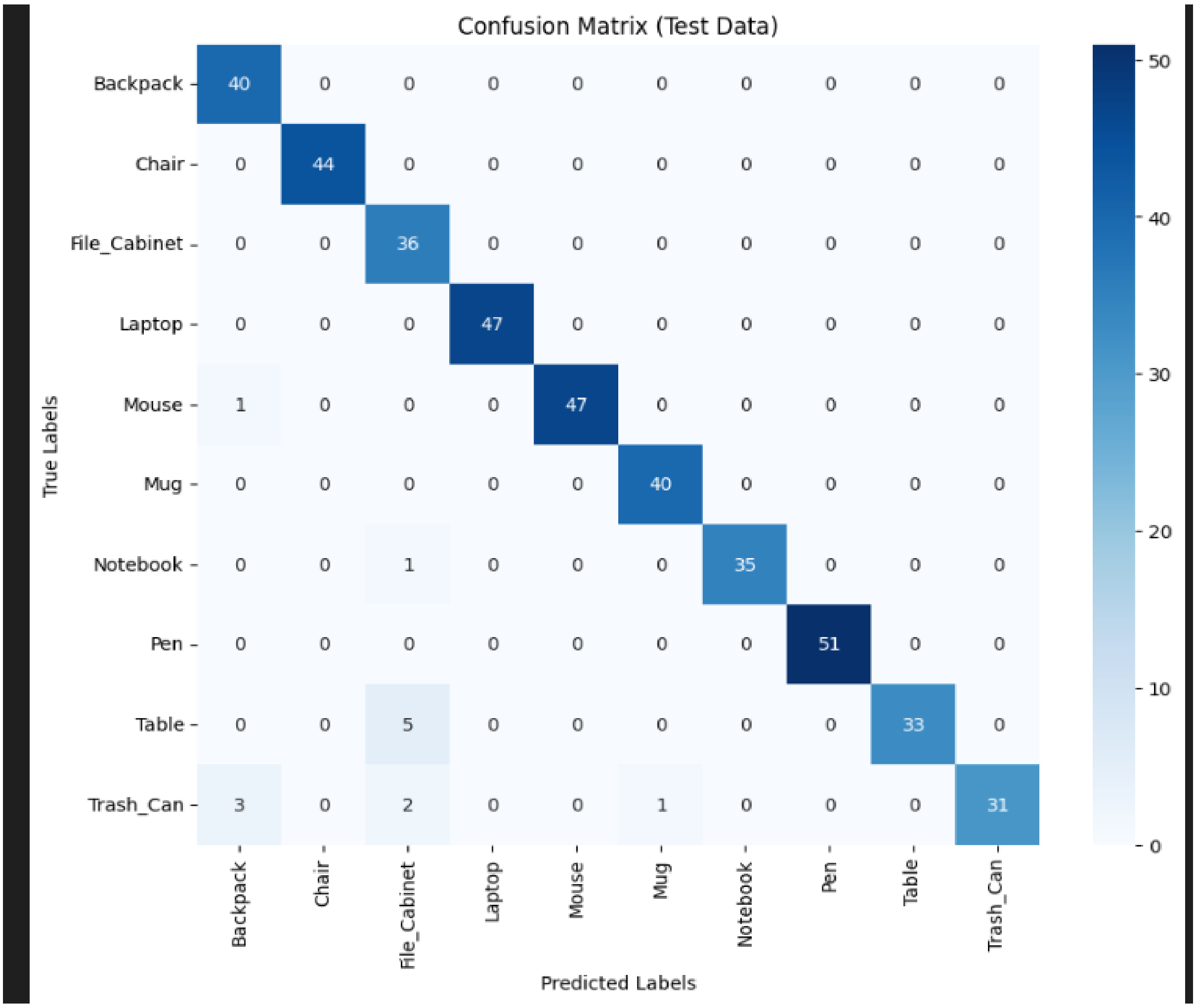
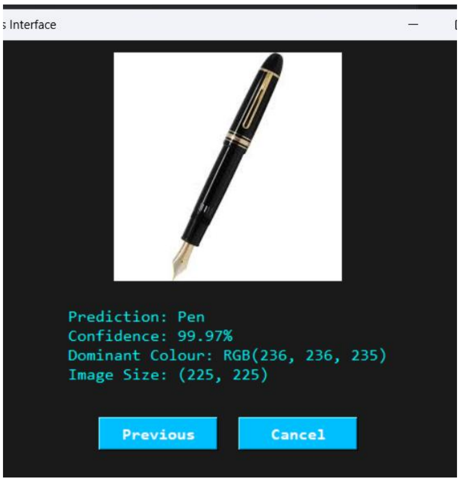


Figure 18 Confusion matrix

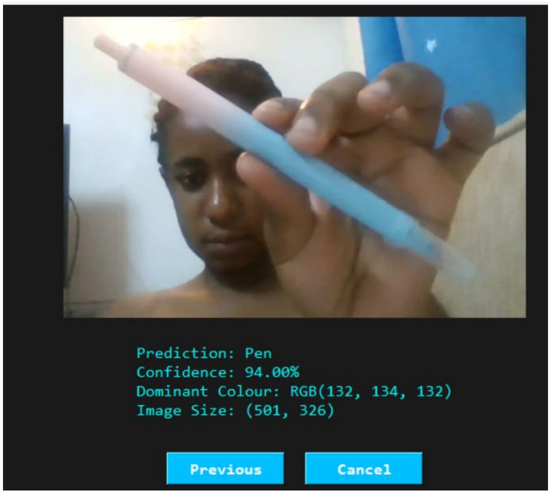
The confusion matrix in figure 9 shows that the model correctly predicted classes most of the time. The highest misclassification was when the model wrongly predicted 5 instances of File cabinet for table, this supports the analysis made above(figure 5) that this probably happened as office tables/desks and file cabinets share same features such as drawers. The number of instances which were wrongly predicted for each class were <=5, which suggests that the model generally performs well and makes a few misclassifications.

Despite the slight overfitting that occurred during training (figure 4), the model still performed very well on the test data(unseen data), with a 96.9% accuracy, an error rate of approximately 3.1% and a number of wrong predictions of instances of each class being <=5 . The overfitting could have led to the small error rate and wrong predictions as the model was not learning but memorizing patterns. The overfitting could have been avoided by reducing the number of epochs, increasing data augmentation or using weight decay to aid in generalisation.

Practically, our classification model generally works well for both the webcam and uploading of an image. Consider the variation below.



Prediction after uploading image



Prediction after capturing from the webcam

Figure 19 Prediction from UI

Here we notice a decrease in confidence between the two interfaces, whereby the webcam predicts lower than the upload. This could be due to the blurriness of the camera and low lighting. In any case, the score are still well fitted with both still performing >90%.

In the context of the Aerius robot, whose main mission is placing office items in their appropriate areas, using a model with 96.9% accuracy and few misclassifications is suitable for deployment. An error rate of approximately 3.1% could lead to certain office items being rarely placed in the wrong areas, which is not life-threatening or of high consequence. Therefore, the model will allow the robot to fulfill its tasks and mission.

Risk and Safety

Emergency Stop (E-stop)

Aerius features an emergency stop (E-stop) that instantly halts all motions when pressed or triggered automatically under abnormal conditions. This fail-safe system complies with ISO 13850 and KUKA Robotics safety guidelines (International Organisation for Standardisation, 2015; KUKA Robotics, 2021).

Speed Caps

Aerius uses software-enforced speed limits to restrict rail and arm movements in shared workspaces. Speed automatically decreases when sensors detect nearby humans. This ensures safe kinetic energy levels in line with ISO/TS 15066 and Universal Robots guidelines (International Organisation for Standardisation, 2016; Universal Robots, 2020).

Minimum Obstacle Distance

Aerius maintains a dynamic safety buffer using sensors and LiDAR to keep a minimum distance of 30–50 cm from people or objects. It slows, stops, or reroutes using its path planner when the buffer is breached. This approach aligns with ISO 10218-2 and Siemens safety guidelines to reduce collision risks (International Organisation for Standardisation, 2011; Siemens, 2022).

Fallback Mode (Detection and Localise Failure)

Aerius enters Fallback Mode when sensors fail or localisation confidence is low, halting motion and returning safely to its dock. This ensures it never operates blindly, following fail-operational design principles. The approach complies with ISO/TR 13309 and NVIDIA safety recommendations (International Organisation for Standardisation, 2018; NVIDIA, 2023).

Proximity & Force Feedback Protection

Aerius uses force and proximity sensors in its arm and soft-grip to detect unexpected resistance or human contact, stopping and retracting instantly. Vision checks ensure safety before resuming motion. This force-feedback system aligns with ISO/TS 15066 and ABB Robotics safety guidelines (International Organisation for Standardisation, 2016; ABB Robotics, 2021).

Thermal, Electrical, and System Safety

Aerius has temperature regulation, current monitoring, and overload protection for its rail and arm actuators. These systems halt operation and send diagnostics if thresholds are exceeded. Such measures ensure electrical safety and device durability, complying with NASA and IEC 60204-1 standards (NASA, 2020; International Electrotechnical Commission, 2016).

Human Presence & Occupancy Detection

Aerius uses AI-based human detection and motion sensors to halt movement near workers, resuming only once the area is clear. This ensures safe operation in shared workspaces. The system complies with ISO 13482:2014 and Boston Dynamics safety guidelines (International Organisation for Standardisation, 2014; Boston Dynamics, 2021).

Below is a safety/fallback decision flow:

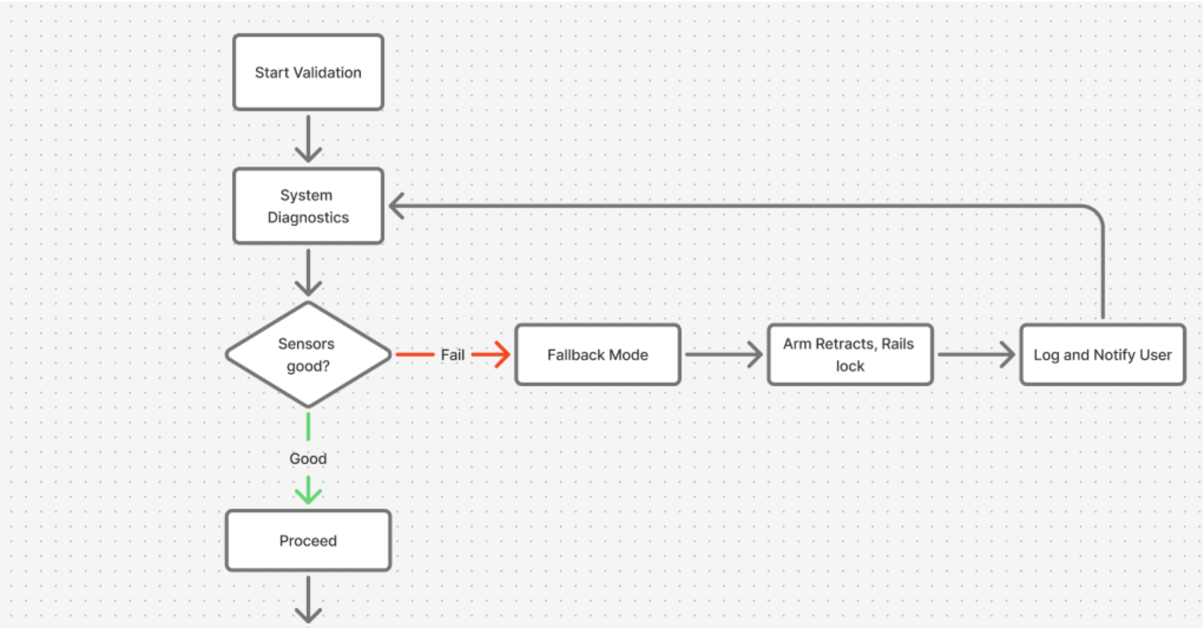


Figure 20- Fallback Decision Flow (1)

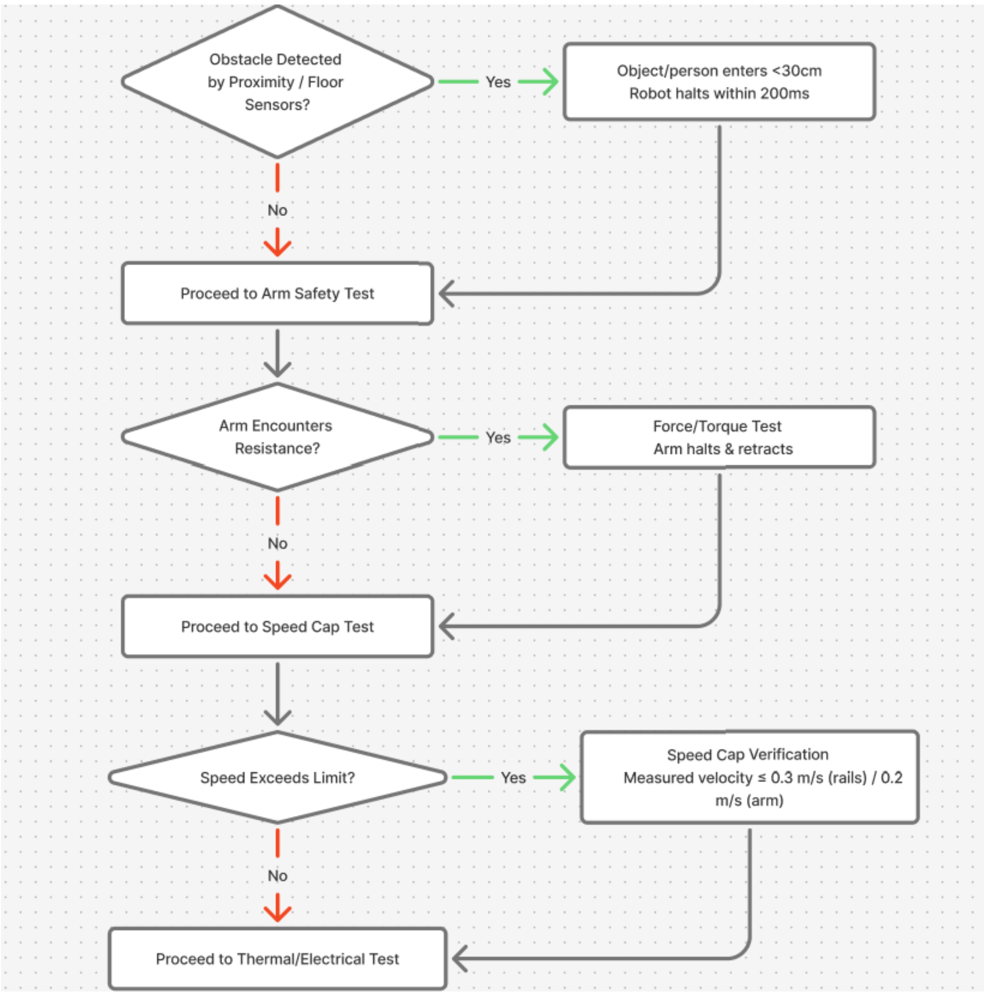


Figure 21 Fallback Decision Flow (2)

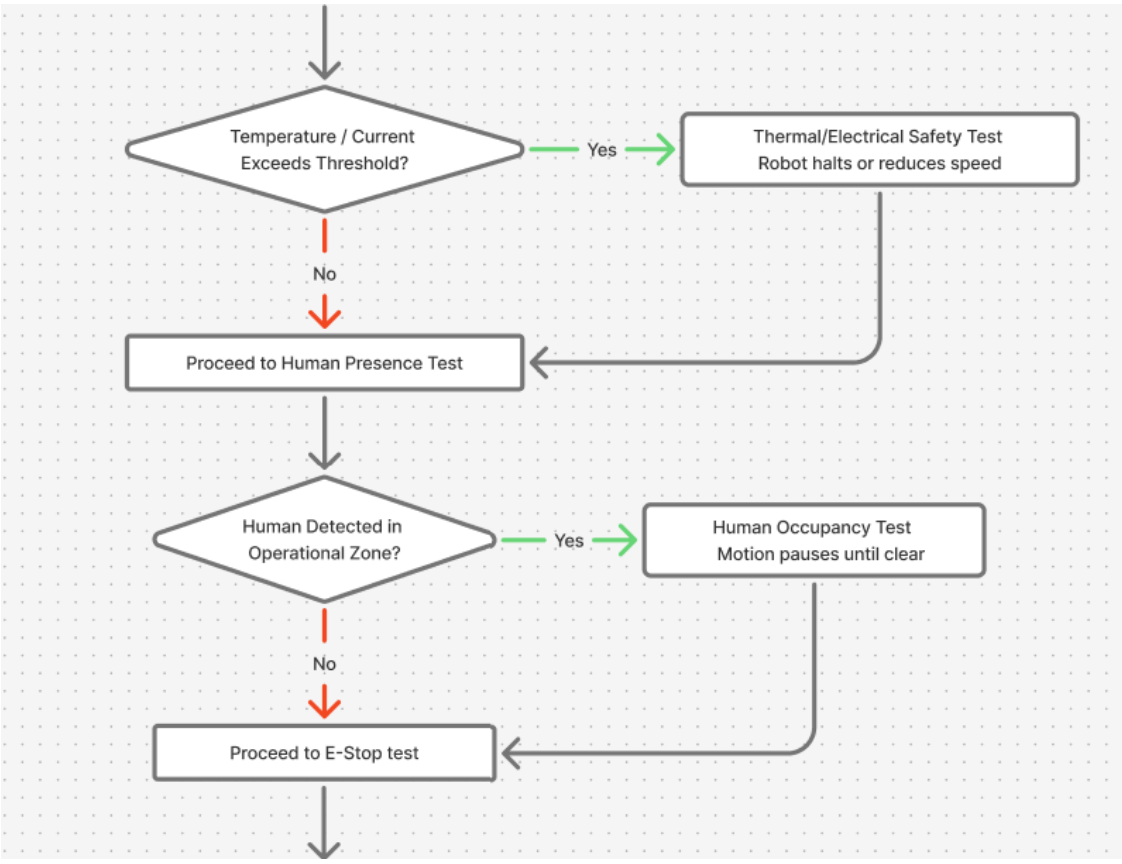


Figure 22 Fallback Decision Flow (3)

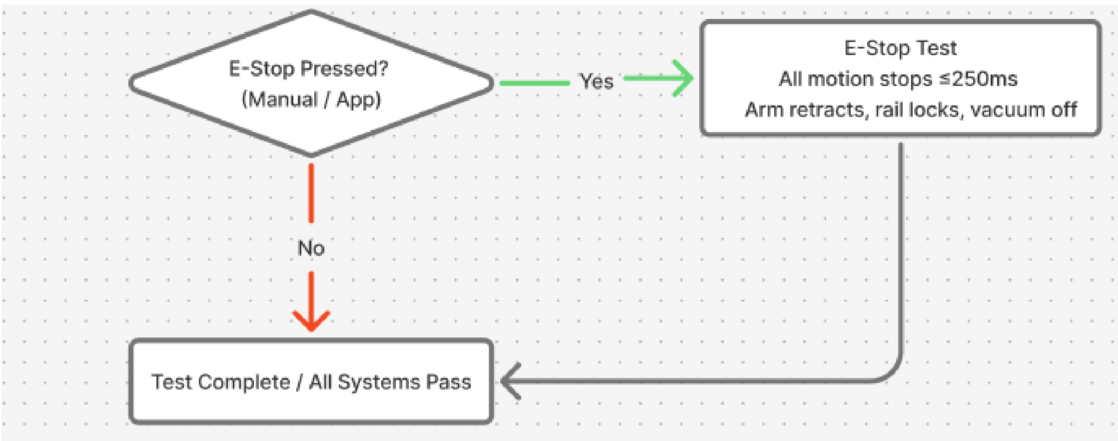


Figure 23 Fallback Decision Flow (4)

Budget / BOM

Allocated Budget**: $6,000**

Estimated Total Cost: **$5,265**

An additional allowance of **$735** is reserved for unexpected expenses, such as price fluctuations, or last-minute component changes. This ensures the project remains within the allocated budget of **$6,000**.

Component	Quantity	Total Price (USD)
Intel RealSense D415 RGB Camera	1	200
OAK-D (Luxonis)	1	350
Slamtec RPLIDAR S2 2D LiDAR	1	400
Bosch BNO055 IMU	1	35
AprilTag v2	1	10
HC-SR04 Safety Proximity Sensor	2	10
UFactory xArm 6 DoF Telescopic Robotic Arm	1	1,200
Electric Linear Actuator	2	300
Robotiq 2F-85 Soft-Grip End Effector	1	400
Stepper Motor Linear Gantry Ceiling Rail Actuators	2	1,600
Trinamic TMC stepper drivers Rail Controller	1	150
Custom Aluminum CNC Ceiling Mount Frame	1	300
ReSpeaker 6-Mic Microphone Array	1	60
2.4" TFT + LED Ring Visual Feedback	1	25
HEPA13 + prefilter Dust Filter	1	50
Air-jet + wiper Sensor Cleaning	1	30
Schneider XB4 Emergency Stop	1	25
ToF + capacitive strips Proximity Sensors	2	120
Total		**5265**

Figure 24 Budget

Conclusion

Aerius is an intelligent ceiling-mounted office assistant that combines cutting-edge sensors, cameras, LiDAR, actuators, algorithms ResNet core with extension of YOLOv8, SLAM , and multi-sensor fusion to accomplish exact item handling, real-time mapping, and precise navigation. Strict testing and well-defined success criteria ensured safety, effectiveness, and flexibility. A dependable and scalable system for an independent office organisation is produced by balancing performance, cost, and integration complexity with the chosen hardware and software solutions.

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Appendix

Contribution Table

Student Name	Student Number	Contribution to Project	Percentage contribution	Signature
Angel Magala	M00971558	Report section: Software architecture. Video. Created Read me file. (GUI of the program) Connected UI to the model. Added evaluation results.	100%	A.M
Mehtaab Andoo	M00889045	GUI of the program (webcam,upload,details) Report sections: Introduction, Mission and task, Required skills, Hardware components, Risks and safety, Budget /BOM. Report compilation. Video	100%	M.A
Sebabatso Maloi	M00919791	Obtained Datasets, created model, trained and tested model Report section: Dataset, model plan, and error analysis. Video	100%	S.M